

Muhammad Suleman

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SUMMARY

AI/ML-focused developer with hands-on experience in Python and machine learning, complemented by a strong software development foundation. Experienced in building practical AI/ML applications involving recommendation systems, data processing, vector similarity, APIs, and interactive ML interfaces. Completed the Machine Learning Specialization and Machine Learning with Python training, with growing expertise in NLP, Generative AI, embeddings, and Retrieval-Augmented Generation (RAG). Seeking an AI/ML Internship to apply machine learning fundamentals and develop practical AI solutions.

EDUCATION

COMSATS University Islamabad — Lahore Campus

Bachelor's in Computer Science, Relevant foundation: Machine Learning, Artificial Intelligence, Object-Oriented Programming, Data Structures & Algorithms, Databases, Software Engineering, and Programming Fundamentals., Programming experience with Python, Java, C, and JavaScript.

PROJECTS

Movie Recommendation System | Python | Scikit-learn | FastAPI | Streamlit

- Built a content-based movie recommendation application using Python and machine learning techniques.
- Processed movie metadata and transformed textual features into numerical representations for recommendation.
- Implemented vector similarity to identify and recommend movies with similar characteristics.
- Developed a FastAPI backend to expose recommendation functionality through APIs.
- Built an interactive Streamlit interface for movie search and recommendation.

RAG-Based Document Question Answering System | Python | Embeddings | LLM | FastAPI | Vector Search

- Developed a Retrieval-Augmented Generation pipeline for answering questions from user-provided documents.
- Implemented document preprocessing, text chunking, embedding generation, and semantic retrieval.
- Used retrieved document context to generate grounded responses through an LLM.
- Built a FastAPI backend for document ingestion and question-answering workflows.
- Designed the system to reduce irrelevant responses by grounding generated answers in retrieved content.

Customer Review Sentiment Analysis | Python | Pandas | Scikit-learn | NLP

- Developed an NLP classification pipeline for identifying positive and negative customer reviews.
- Applied text preprocessing and TF-IDF vectorization to convert textual data into machine-learning features.
- Trained classification models and evaluated performance using accuracy, precision, recall, and F1-score.
- Used confusion-matrix analysis to identify classification errors and evaluate model performance.

AI Resume–Job Matching System | Python | NLP | Embeddings | FastAPI

- Developed an NLP-based system for matching candidate resumes against job descriptions.
- Extracted relevant skills and represented resume and job-description content using semantic embeddings.
- Calculated similarity scores to rank resumes according to their relevance to a target position.
- Exposed the matching pipeline through a FastAPI service.

TECHNICAL SKILLS

Programming: Python, SQL, Java, C, JavaScript

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Feature Engineering, Model Evaluation, Recommendation Systems

AI / NLP: Natural Language Processing, Text Classification, TF-IDF, Vector Similarity, Embeddings, Prompt Engineering, Generative AI, RAG

Python Libraries: NumPy, Pandas, Scikit-learn, Matplotlib

AI Development: FastAPI, Streamlit, Jupyter Notebook

Databases: PostgreSQL, MySQL, MongoDB, Supabase

Tools: Git, GitHub, Docker, Postman

CERTIFICATIONS

[Machine Learning Specialization](#) — Machine Learning fundamentals and practical model development

[Machine Learning with Python](#) — Python, Machine Learning Algorithms, Supervised Learning, Unsupervised Learning

[Introduction to Software Engineering](#) — SDLC, Software Architecture, Design Patterns, Python, Development Tools

[IBM Front-End Developer Professional Certificate](#) — Modern frontend development and production-ready development workflows